University of St.Gallen

Stochastic Modeling in Finance and Insurance

Chapter 4: Interest Rate Risk

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"From insight to impact"

Learning goals for chapter 4

After this chapter, you will be able to…

- distinguish the risk-neutral measure from the forward measure for interest rate valuation.
- interpret forward prices as expectations of future spot prices under the forward measure.
- fit the Nelson-Siegel-Svensson model to observed spot rates and forward rates.
- simulate the Hull-White short rate model and derive affine term structures.

Interest Rate Risk

Interest Rate Risk

Introduction

Definition

Interest rate risk is the risk of fluctuating interest rates.

- Common measures for the interest rate sensitivity of financial instruments: Duration, PV01 (PVBP)
- Entirely systematic (shifts in the term structure affect the whole market)
- Can be hedged with interest rate derivatives (forwards/futures, swaps, caps/floors, swaptions)

Occurrence

- Finance: Fixed income mutual funds, hedge funds, pension funds, (investment) banks
- Insurance: Asset-liability management of (life) insurance companies

Interest Rate Risk

Some particularities

Differences to equity risk

Interest rate risk is more difficult to model than equity risk.

- Individual interest rates are less well behaved than stock prices/returns.
- Most products require a model for the entire yield curve.
- In many cases, interest rates are needed for both payoffs and discounting.

When do we need a stochastic term structure model?

- Bonds and vanilla swaps can be priced under the *forward measure.*
- Bond options, caps/floors, and swaptions can be priced with the *Black model.*
- Stochastic term structure models are required for exotic derivatives and for risk management purposes!

Interest Rate Risk

The term structure of interest rates

Definition

The term structure (yield curve) is a representation of the relationship between market remuneration rates and the remaining time to maturity of debt securities.

Estimation

Central banks estimate (zero-coupon) spot and (instantaneous) forward curves for their currency area.

- Most yields are unobservable and must be estimated from existing zero bonds and fixed coupon bonds.
- A mathematical function is fit to the raw yield data, using a yield curve model (e.g., Nelson-Siegel-Svensson).
- Yield curves reflect continuous (log) rates.

Source: European Central Bank.

Interest Rate Risk

Complications for risk-neutral valuation

Problem 1

Under traditional risk-neutral valuation, we calculate the expected future payoff by assuming that the expected return on the underlying asset equals the constant risk-free rate (see GBM drift in the B-S model).

► Interest rate risk – by definition – requires rates to be *stochastic* → drift changes along the return path

Problem 2

The concept of the risk-free rate is not clear cut in the presence of a stochastic term structure.

- One-day risk-free rate as the expected return for daily time steps?
- One-year risk-free rate as the expected return for yearly time steps?
- Current 5-year rate as the expected return for the asset price distribution after a 5-year period?

► The risk-neutral measure is no longer unique.

Interest Rate Risk

Risk-neutral valuation with stochastic interest rates

Consider the evolution of a money market account $\widetilde{M}$ according to an *instantaneous* risk-free rate $\tilde{r}$:

$$\widetilde{M}_T = M_0 \exp\left(\int_0^T \tilde{r}\,\mathrm{d}t\right) \quad \rightarrow \quad P_0 = \mathbb{E}^{\mathbb{Q}_*}\left(\exp\left(-\int_0^T \tilde{r}\,\mathrm{d}t\right)\tilde{P}_T\right) = \mathbb{E}^{\mathbb{Q}_*}\left(\exp(-\bar{r}T)\,\tilde{P}_T\right)$$

- $\tilde{r}$ is called *stochastic short rate* → $\bar{r}$ is the average value of $\tilde{r}$ between $t = 0$ and $t = T$,
- P_0 is the price of a risky asset with payoff $\tilde{P}_T$,
- $\mathbb{E}^{\mathbb{Q}_*}$ is the expectation under the (traditional) *risk-neutral measure*.

► One way of pricing interest-rate sensitive assets is to simulate $\tilde{r}$ in a risk-neutral world. For each trial, the asset payoff is calculated and discounted at the average value of $\tilde{r}$ on the <u>sampled path</u>.

Note: When rates are assumed to be constant, the expression above reduces to: $P_0 = \exp(-rT)\,\mathbb{E}^{\mathbb{Q}}(P_T)$.

Interest Rate Risk

From the risk-neutral measure to the forward measure

Consider a zero-coupon bond (with maturity T) as the numeraire (rather than a money market account):

$$P_0 = DF(0,T) \cdot \mathbb{E}^{\mathbb{Q}_T}\left(\tilde{P}_T\right)$$

- $DF(0,T)$ is the discount factor for the period between $t = 0$ and $t = T$ from the current term structure,
- P_0 is the price of a risky asset with payoff $\tilde{P}_T$,
- $\mathbb{E}^{\mathbb{Q}_T}$ is the expectation under the *forward measure*.

► Another way of pricing interest-rate sensitive assets is to form an expectation about their payoff at time T and discount it with the interest rate for maturity T from the current term structure.

Note: The discount factor is now outside the expectations operator!

Interest Rate Risk

Forward prices as expectations of future spot prices

Consider a risky asset with current price P_0 and stochastic future price $\tilde{P}_T$ (at time T).

A forward contract on this asset with maturity T is defined as a contract that pays off $\tilde{P}_T - X$ at time T, with X being the delivery price. Define V_0 as the present value of this forward contract:

$$V_0 = DF(0,T) \cdot \mathbb{E}^{\mathbb{Q}_T}\left(\tilde{P}_T - X\right).$$

The fair forward price $F_0(T)$ of the asset at time 0 is the value of X for which V_0 equals zero. It follows that

$$DF(0,T)\left(\mathbb{E}^{\mathbb{Q}_T}\left(\tilde{P}_T\right) - F_0(T)\right) = 0$$

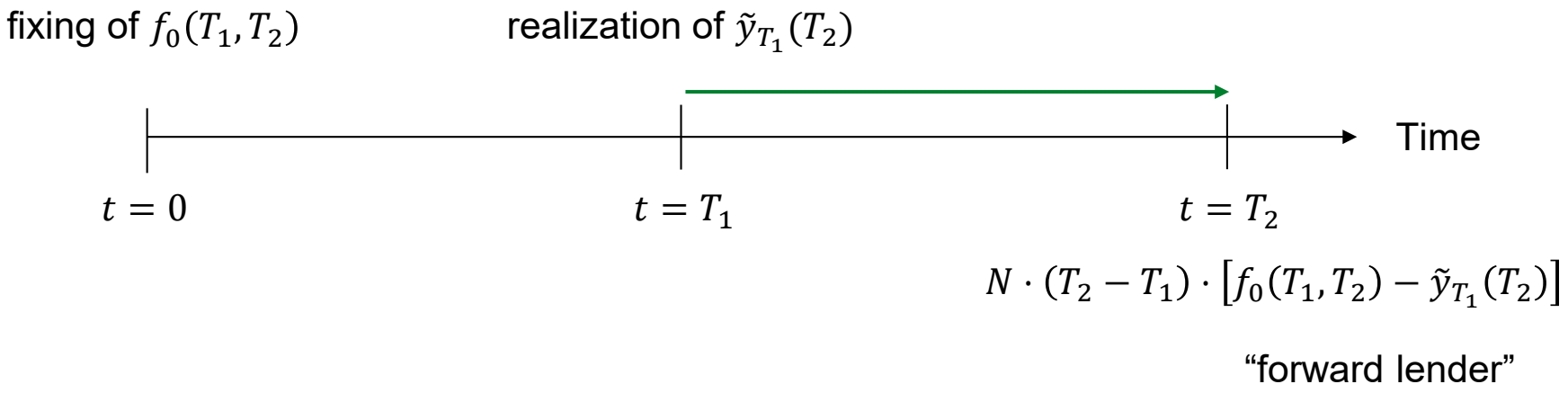
$$F_0(T) = \mathbb{E}^{\mathbb{Q}_T}\left(\tilde{P}_T\right).$$

- Under the forward measure $\mathbb{Q}_T$, the forward price equals the expected future spot price.

Interest Rate Risk

Forward Rate Agreements (FRAs)

A *forward rate agreement* (FRA) with maturity T_1 and notional N pays off based on the difference between a forward rate $f_0(T_1, T_2)$ fixed today and a future realized interest rate $\tilde{y}_{T_1}(T_2)$ for the period from T_1 to T_2.



- T_1 is called the effective date and T_2 is called the termination date ($T_1 < T_2$).
- A FRA is a contract for difference (no actual investment/borrowing).
- Settlement is at T_1 (payoff discounted from T_2 to T_1, using the realized spot rate $\tilde{y}_{T_1}(T_2)$).

Interest Rate Risk

Forward rates as expectations of future spot rates

Define V_0 as the present value of a FRA (under the forward measure):

$$V_0 = DF(0,T_2) \cdot N \cdot (T_2 - T_1) \cdot \mathbb{E}^{\mathbb{Q}_{T_2}} \left(f_0(T_1,T_2) - \tilde{y}_{T_1}(T_2) \right).$$

The fair forward rate $f_0(T_1,T_2)$ at time $t = 0$ ensures that V_0 equals zero:

$$DF(0,T_2) \cdot N \cdot (T_2 - T_1) \left(f_0(T_1,T_2) - \mathbb{E}^{\mathbb{Q}_{T_2}} \left(\tilde{y}_{T_1}(T_2) \right) \right) = 0$$

$$f_0(T_1,T_2) = \mathbb{E}^{\mathbb{Q}_{T_2}} \left(\tilde{y}_{T_1}(T_2) \right).$$

- Under the forward measure $\mathbb{Q}_{T_2}$, the forward rate equals the expected future spot rate.

The Nelson-Siegel-Svensson Model

The Nelson-Siegel-Svensson Model

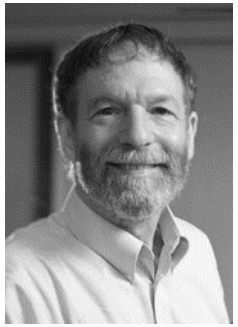
Assumptions and input parameters

Assumptions

- No specific economic assumptions (pure econometric model)
- Mathematical function chosen to maximize flexibility
  - ► Model captures important attributes of the yield-maturity relation
  - ► Fits the typical yield curve shapes observed in practice

Input parameters

- Level of the yield curve β_0
- Slope of the yield curve β_1
- Curvature of the yield curve (hump) β_2, β_3
- Decay rates (hump) λ_1, λ_2



Charles R. Nelson & Andrew F. Siegel
"Parsimonious Modeling of Yield Curves"
Journal of Business, October 1987

extended by
Lars E. O. Svensson (1994)

The Nelson-Siegel-Svensson Model

Model specification for the spot rate curve

Parametric yield curve models specify a function that is defined over the entire maturity domain.
The *Nelson-Siegel-Svensson* model for the spot interest rate curve $y_0(T)$ looks as follows:

$$y_0(T) = \beta_0 + \beta_1 \left(\frac{1 - \exp(-\lambda_1 T)}{\lambda_1 T} \right) + \beta_2 \left(\frac{1 - \exp(-\lambda_1 T)}{\lambda_1 T} - \exp(-\lambda_1 T) \right) + \beta_3 \left(\frac{1 - \exp(-\lambda_2 T)}{\lambda_2 T} - \exp(-\lambda_2 T) \right)$$

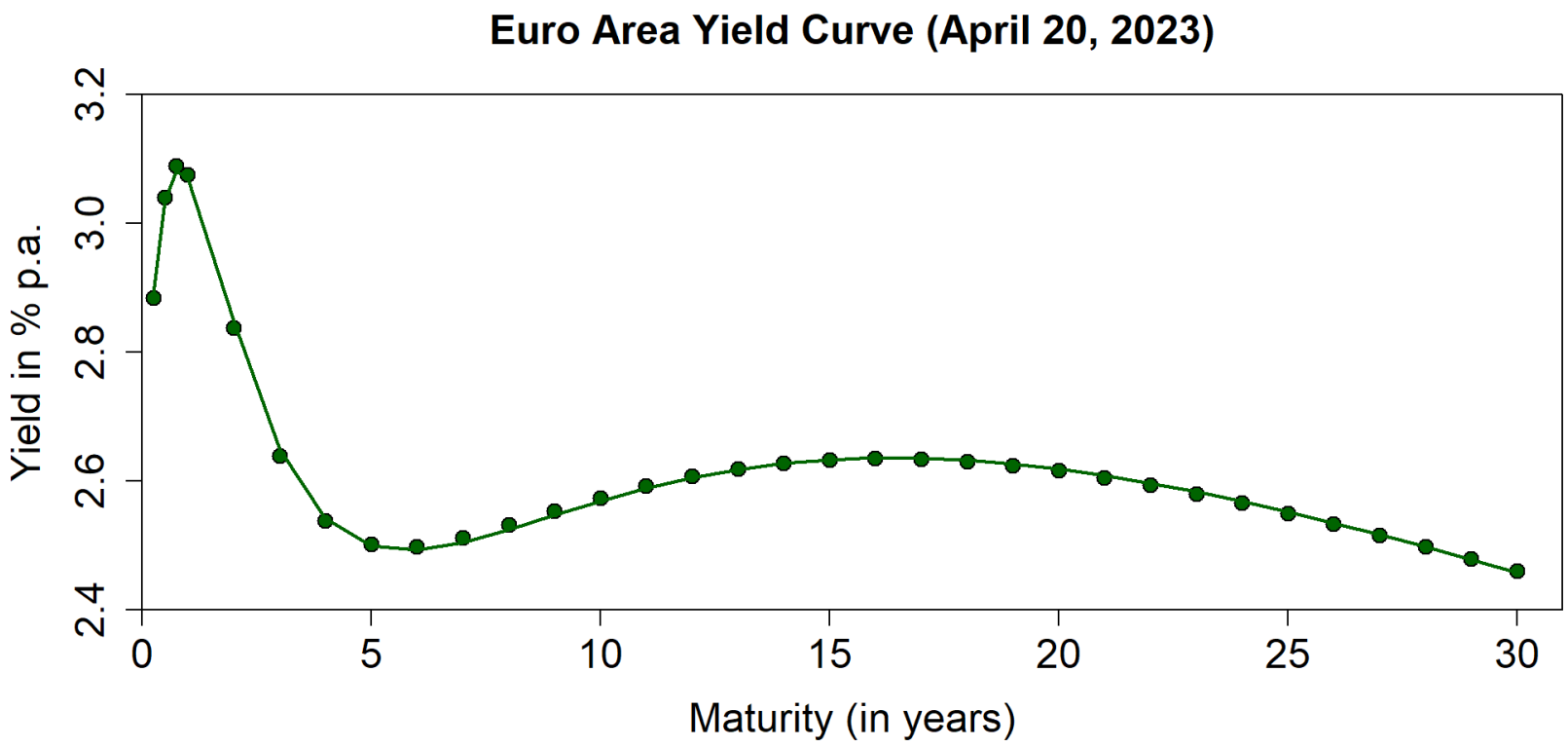
- T denotes the maturity of the zero-coupon bonds corresponding to the yields,
- $\beta_0, \beta_1, \beta_2, \beta_3, \lambda_1, \lambda_2$: parameters to be estimated (level, slope, curvature, decay).
  (estimation: minimization of the squared deviations of theoretical prices from observed prices)

► The Nelson-Siegel-Svensson model is used by many central banks around the world.*)

*) An alternative are spline approaches, which rely on a piecewise polynomial instead of a single functional form over the entire maturity range.

The Nelson-Siegel-Svensson Model

Illustration of the spot yield curve

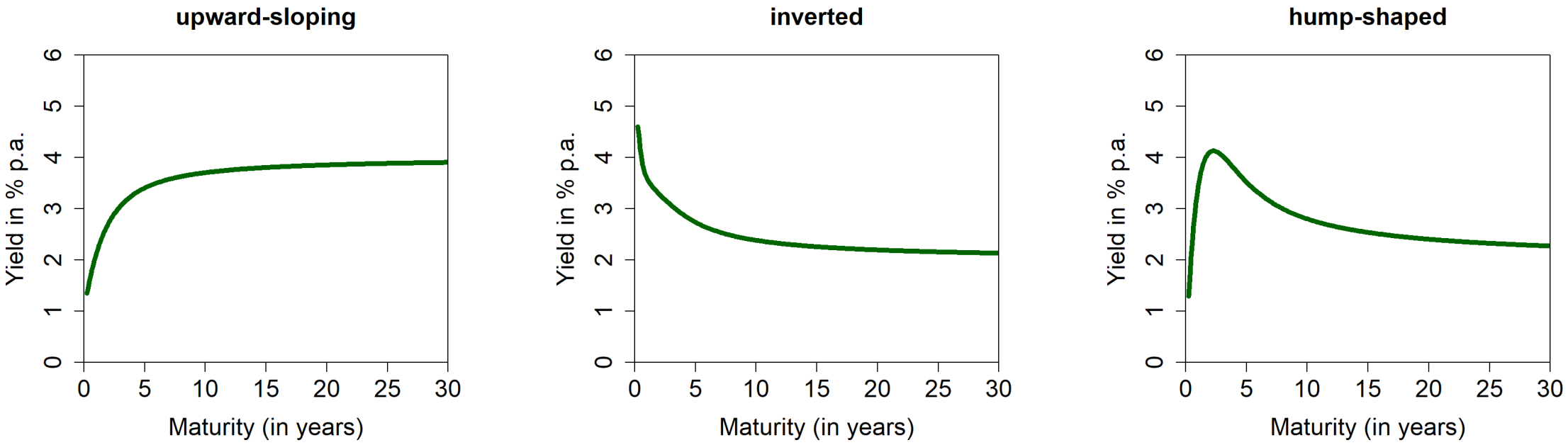


Source: European Central Bank.

The Nelson-Siegel-Svensson Model

Illustration of yield curve shapes

The model can produce upward-sloping, inverted and hump-shaped yield curves



Parameters: see R code.

The Nelson-Siegel-Svensson Model

Extracting arbitrage-free forward rates from the yield curve

Define $y_0(T)$ as the current spot rate for maturity T and $f_0(T_1, T_2)$ as the forward rate fixed today for the future period from T_1 to T_2. To rule out arbitrage, the following relation must hold:

$$\exp(y_0(T_2)T_2) = \exp(y_0(T_1)T_1) \cdot \exp\big(f_0(T_1, T_2)(T_2 - T_1)\big).$$

Take logs:

$$y_0(T_2)T_2 = y_0(T_1)T_1 + f_0(T_1, T_2)(T_2 - T_1)$$

$$f_0(T_1, T_2) = \frac{y_0(T_2)T_2 - y_0(T_1)T_1}{T_2 - T_1}.$$

- Arbitrage-free forward rates can be extracted from the current spot yield curve.

The Nelson-Siegel-Svensson Model

Model specification for the instantaneous forward curve

When the period between effective date and termination date tends to zero, we obtain the *instantaneous forward rate* $f_0(T, T + \mathrm{d}t)$. The Nelson-Siegel-Svensson model for $f_0(T, T + \mathrm{d}t)$, looks as follows:

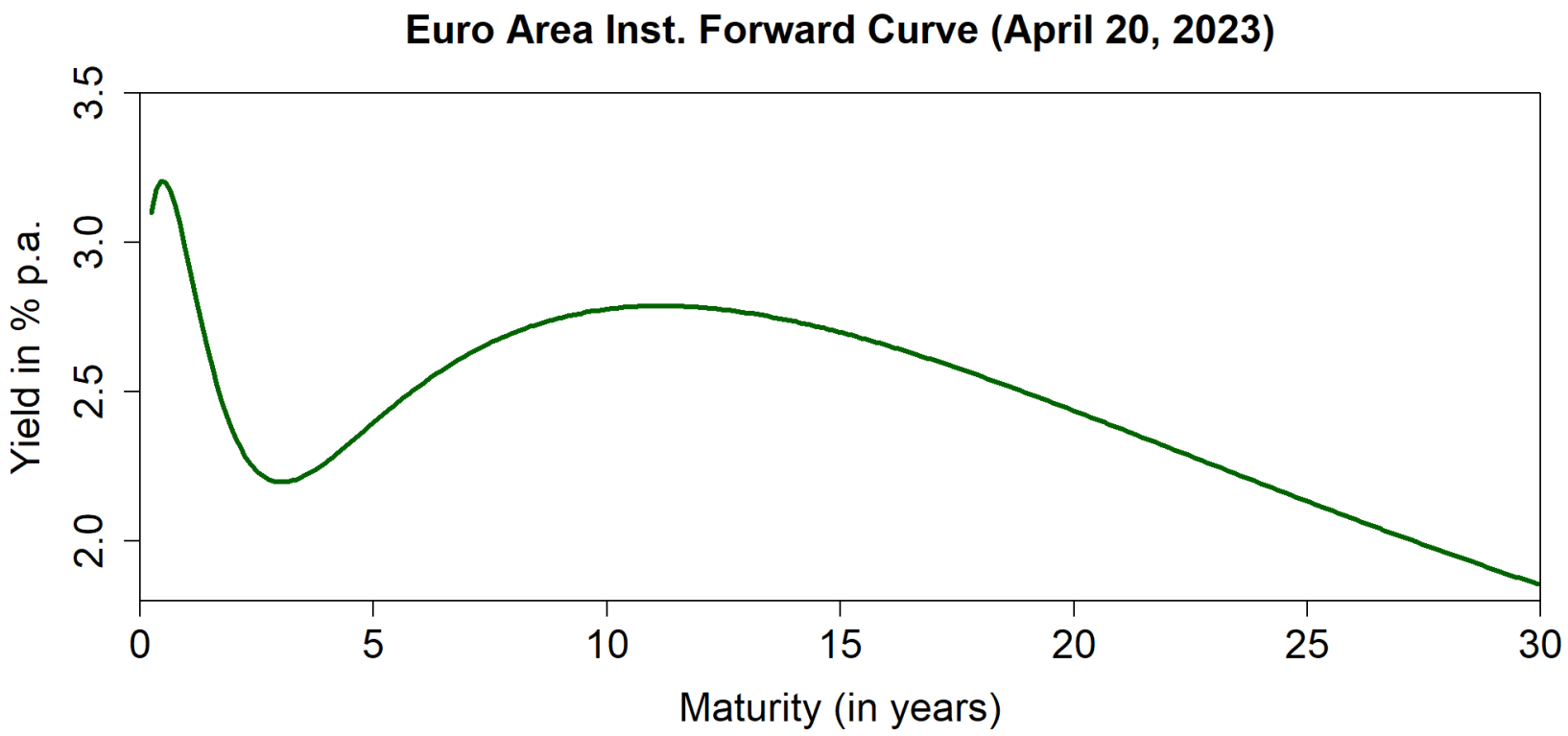
$$f_0(T, T + \mathrm{d}t) = \beta_0 + \beta_1 \exp(-\lambda_1 T) + \beta_2\, \lambda_1 T \exp(-\lambda_1 T) + \beta_3\, \lambda_2 T \exp(-\lambda_2 T)$$

- T denotes the maturity of the (hypothetical) FRAs,
- $\beta_0, \beta_1, \beta_2, \beta_3, \lambda_1, \lambda_2$: parameters to be estimated (level, slope, curvature, decay rate).
  (estimation: minimization of the squared deviations of theoretical prices from observed prices)

► The instantaneous forward rate is a key input in the calibration of the Hull-White model.

The Nelson-Siegel-Svensson Model

Illustration of the Instantaneous forward curve



Source: European Central Bank.

The Hull-White Model

The Hull-White Model

Typology of stochastic term structure models

	Equilibrium models	No-arbitrage models
Synopsis	► Today's term structure is an output ► No exact fit (approximation) ► Suitable for long-term risk analysis (life insurance, pension funds)	► Today's term structure is an input ► Exact fit (no pricing error) ► Suitable for derivatives pricing (investment banks, hedge funds)
One Factor	Vasicek Model (1977) Cox-Ingersoll-Ross Model (1985)	Ho-Lee Model (1986) Hull-White Model (1990)
Two Factors	Longstaff-Schwartz Model (1992)	2-Factor Hull-White Model (1994)

The Hull-White Model

General logic of stochastic short-rate models

Consider the time-t price $\tilde{P}_t(T)$ of a zero-coupon bond with maturity T that pays back \$1:

$$\tilde{P}_t(T) = \mathbb{E}^{\mathbb{Q}_*}(\exp(-\bar{r}(T-t))\,1) \tag{1}$$

where $\bar{r}$ is the average value of the stochastic instantaneous short rate $\tilde{r}$ between t and T.
Denoting $\tilde{y}_t(T)$ as the continuously-compounded risk-free rate at time t for maturity T, we may write

$$\tilde{P}_t(T) = \exp\big(-\tilde{y}_t(T)(T-t)\big) \qquad \rightarrow \qquad \tilde{y}_t(T) = -\frac{\ln \tilde{P}_t(T)}{T-t}. \tag{2}$$

Combining (1) and (2) delivers:

$$\tilde{y}_t(T) = -\frac{\ln \mathbb{E}^{\mathbb{Q}_*}\big(\exp\big(-\bar{r}(T-t)\big)\big)}{T-t}.$$

- The term structure of interest rates at any time t can be obtained from the (risk-neutral) process of $\tilde{r}$. Once this process has been defined, it links the initial zero curve and its evolution through time.

The Hull-White Model

Assumptions and input parameters

Assumptions

- Arbitrage-free markets without frictions (transaction costs, taxes etc.)
- Interest rates follow a mean-reverting process
- Mean reversion level is dependent on time
- Constant short rate volatility

Input parameters

- Current term structure (or zero bond prices) $y_0(T)$ or $P_0(T)$
- Volatility (standard deviation) of the short rate σ
- Mean-reversion speed α

  (level θ_t is determined by the initial term structure)

John C. Hull &
Alan D. White
"Pricing Interest Rate Derivative Securities"
Review of Financial Studies, October 1990

The Hull-White Model

Short rate process

Hull and White (1990) describe the dynamics of the short rate $\tilde{r}_t$ by an Ornstein-Uhlenbeck process:

$$\mathrm{d}\tilde{r}_t = \alpha\left(\frac{\theta_t}{\alpha} - r_t\right)\mathrm{d}t + \sigma\,\mathrm{d}\widetilde{W}_t$$

- α (mean reversion speed) and σ (volatility) are constants,
- The mean reversion level θ_t/α is time dependent,
- $\mathrm{d}\widetilde{W}_t$ is a continuous-time Wiener process.

Nomenclature

In their original paper, Hull and White (1990) allow for the mean reversion rate α and the volatility factor σ to be functions of time as well, calling their model the "*extended Vasicek model*". The term "*Hull-White model*" has later been coined by practitioners for the short rate process above, in which only θ_t is time-dependent.

The Hull-White Model

Analytical tractability

θ_t is a function of time, chosen to ensure that the model fits the initial term structure:

$$\theta_t = \frac{\partial f_0(t, t + \mathrm{d}t)}{\partial t} + \alpha f_0(t, t + \mathrm{d}t) + \frac{\sigma^2}{2\alpha}(1 - \exp(-2\alpha t))$$

where $f_0(t, t + dt)$ is the *instantaneous forward rate* for maturity t as seen at time $t = 0$.
The last term is usually very small. Ignoring it implies a time-varying drift of:*)

$$\alpha\left(\frac{\theta_t}{\alpha} - r_t\right) = (\theta_t - \alpha r_t) = \frac{\partial f_0(t, t + \mathrm{d}t)}{\partial t} + \alpha(f_0(t, t + \mathrm{d}t) - r_t).$$

► Thus, on average, $\tilde{r}_t$ follows the instantaneous forward rate curve.
After deviations from that curve, it reverts back to it at rate α.

*) Based on the Nelson-Siegel-Svensson model, the partial derivative equals: $f_0'(t, t + \mathrm{d}t) = -\lambda_1 \exp(-\lambda_1 t)(\beta_1 - \beta_2 + \beta_2 \lambda_1 t) + \lambda_2 \exp(-\lambda_2 t)(\beta_3 - \beta_3 \lambda_2 t)$.

The Hull-White Model

Distribution of the short rate

Solving the SDE delivers the variable $\tilde{r}_t$:

$$\tilde{r}_t = r_0 \exp(-\alpha t) + \int_0^t \exp\big(\alpha(s-t)\big)\,\theta(s)\,\mathrm{d}s + \sigma \exp(-\alpha t) \int_0^t \exp(\alpha u)\,\mathrm{d}\widetilde{W}(u).$$

► Since $\mathrm{d}\widetilde{W}_t$ is a normally-distributed random variable, $\tilde{r}_t$ is also a normally-distributed random variable with

$$\mathbb{E}[\tilde{r}_t] = r_0 \exp(-\alpha t) + \int_0^t \exp\big(\alpha(s-t)\big)\,\theta(s)\,\mathrm{d}s,$$

$$\mathrm{Var}[\tilde{r}_t] = \frac{\sigma^2}{2\alpha}(1 - \exp(-2\alpha t)).$$

The Hull-White Model

Euler discretization for Monte Carlo simulation

Euler discretization of the Hull-White process with step size Δt delivers:*)

$$\tilde{r}_t = r_{t-1} + (\theta_{t-1} - \alpha r_{t-1})\,\Delta t + \sigma\,\Delta\widetilde{W}_t$$

with time-varying mean-reversion level and discretized Wiener process:

$$\theta_{t-1} = f_0'(t-\Delta t, t) + \alpha f_0(t-\Delta t, t)$$

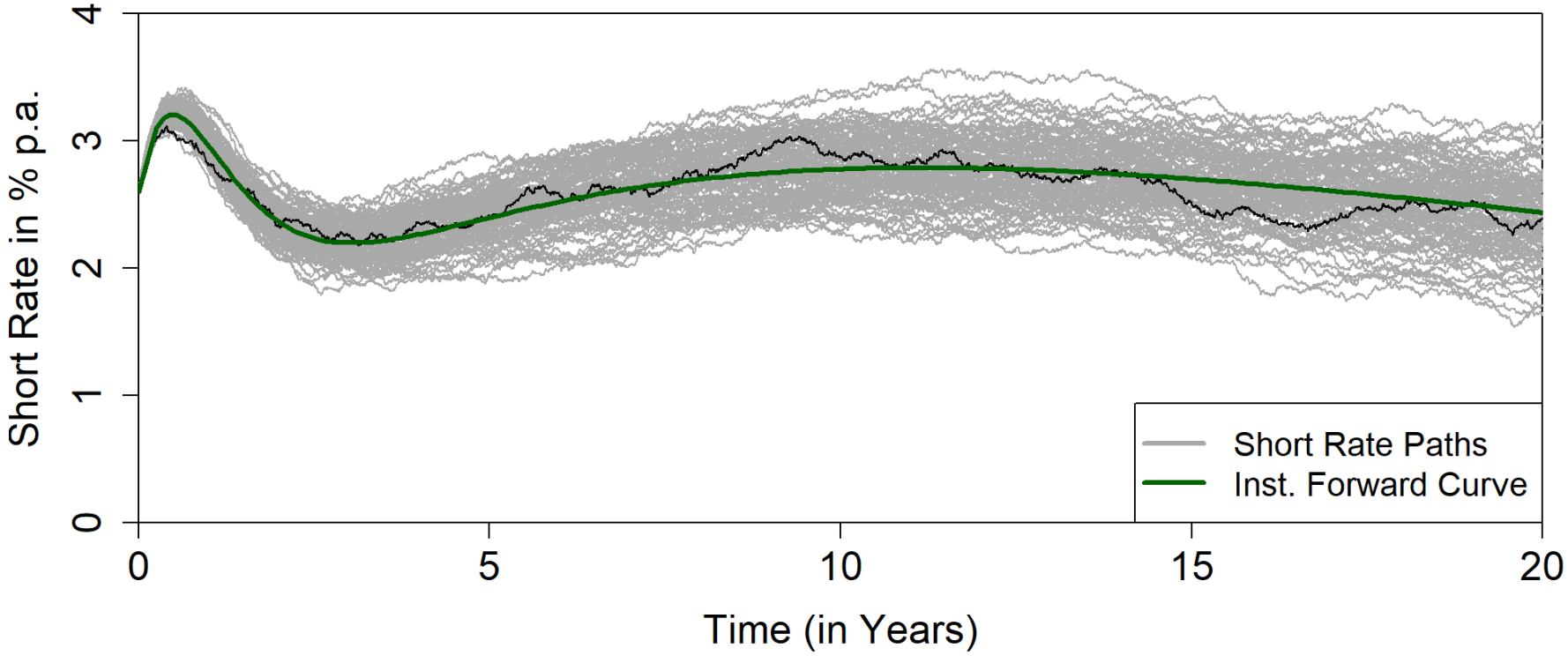
$$\Delta\widetilde{W}_t \sim N\left(0, \sqrt{\Delta t}\right)$$

► The instantaneous forward rate is a key input in the calibration of the Hull-White model.

*) **Note:** the $t-1$ in r_{t-1} is a counting index. It does not imply a step size of 1!

The Hull-White Model

Sample paths



Calibration: Euro Area Yield Curve (April 20, 2023), further parameters see R Code.

The Hull-White Model

Future zero-coupon bond prices

It can be shown that the time-t zero-coupon bond prices in the Hull-White model are given by

$$\tilde{P}_t(T) = A(t,T) \cdot \exp(-B(t,T) \cdot \tilde{r}_t) \quad \rightarrow \quad \ln \tilde{P}_t(T) = \ln A(t,T) - B(t,T) \cdot \tilde{r}_t$$

where $T > \mathrm{t}$,

$$B(t,T) = \frac{1 - \exp\big(-\alpha(T-t)\big)}{\alpha},$$

and

$$\ln A(t,T) = \ln \frac{P_0(T)}{P_0(t)} + B(t,T) \cdot f_0(t, t + \mathrm{d}t) - \frac{\sigma^2}{4\alpha}(1 - \exp(-2\alpha t))\, B(t,T)^2.$$

- The Hull-White model is an *affine term structure model*: any **log** bond price at a future time t is a linear function of the short rate at time t (and today's zero bond prices).

Note: $f_0(t, t + \mathrm{d}t) = -\frac{\partial \ln P_0(t)}{\partial t}$.

The Hull-White Model

Future term structures

The zero bond prices are the flip side of the term structure.
To see this, express them as a function of the yield:

$$\tilde{P}_t(T) = \exp(-\tilde{y}_t(T)(T-t)) \quad \rightarrow \quad \tilde{y}_t(T) = -\frac{\ln \tilde{P}_t(T)}{T-t}.$$

Then insert

$$\ln \tilde{P}_t(T) = \ln A(t,T) - B(t,T) \cdot \tilde{r}_t$$

to obtain:

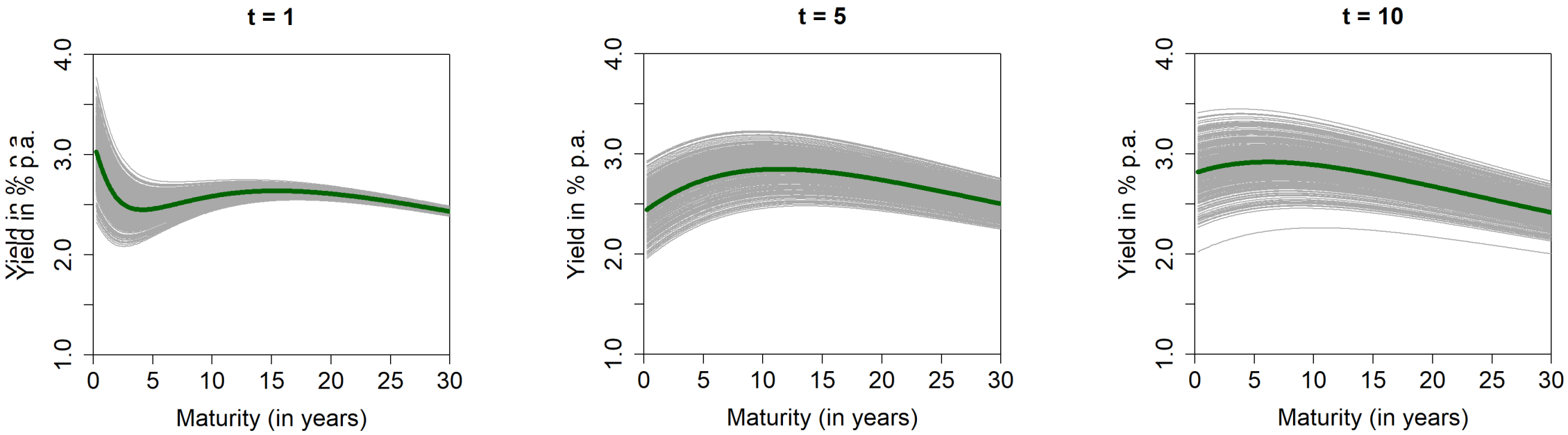
$$\tilde{y}_t(T) = -\frac{\ln A(t,T)}{T-t} + \frac{B(t,T)}{T-t} \cdot \tilde{r}_t.$$

► The term structure at a future time t is a linear function of the short rate at time t.

Note: Here T is defined as a point in time. The relation to the time-to-maturity of a zero bond at time t, m_t (e.g., 5 years), is as follows: $T = t + m_t$.

The Hull-White Model

Average realized term structures



Calibration: Euro Area Yield Curve (April 20, 2023), further parameters see R Code.

The Hull-White Model

Critical reflection

Advantages

- Analytical tractability (closed-form zero-coupon bond prices / yields)
- Can produce negative interest rates (formerly this was seen as a disadvantage)
- $f_0(t, t + \mathrm{d}t)$ and r_0 can be extracted from the current yield curve
- α and σ are determined via calibration to the prices of traded caplets and swaptions

Drawbacks

- Single-factor models imply rigid yield curve movements.
- Conditional on time t, long-term yields are deterministic mappings of the short rate (see slide 31).

Literature

Required literature

Lecture Notes

Recommended texts

Ang, C. S. (2021), Analyzing Financial Data and Implementing Financial Models Using R, 2nd Edition, Springer, ISBN: 978-3-030-64154-2. Chapter 9.

Black, F. (1976), The Pricing of Commodity Contracts, *Journal of Financial Economics*, 3(1−2), 167−179.

Cox, J. C., Ingersoll, J. E. and Ross, S. A. (1985), A Theory of the Term Structure of Interest Rates, *Econometrica*, 53(2), 385−407.

Ho, S. Y. and Lee, S.-B. (1986), Term Structure Movements and Pricing Interest Rate Contingent Claims, *Journal of Finance*, 41(5), 1011−1029.

Hull, J. C. (2015), Options, Futures and Other Derivatives, 11th Edition, ISBN: 978-0-133-45631-8. Chapters 29, 31 and 32.

Literature

Recommended texts (continued)

Hull, J. C. and White, A. D. (1990), Pricing Interest-Rate-Derivative Securities, *Review of Financial Studies*, 3(4), 573−592.

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Longstaff, F. A. and Schwartz, E.S. (1992), Interest Rate Volatility and the Term Structure: A Two-Factor General Equilibrium Model, *Journal of Finance*, 47(4), 1259−1282.

Nelson, C. R. and Siegel, A. F. (1987), Parsimonious Modeling of Yield Curves, *Journal of Business,* 60(4), 473−489.

Svensson, L. E. O. (1994), Estimating and Interpreting Forward Interest Rates: Sweden 1992-1994, *IMF Working Paper*, WP/94/114.

Vasicek, O. (1977), An Equilibrium Characterization of the Term Structure, *Journal of Financial Economics*, 5(2), 177−188.

Appendix:
The Black Model

The Black Model

European bond options

Problem

The Black-Scholes model is not applicable.

- ► Bonds (the underlying) exhibit a pull-to-par effect.
- ► As the bond nears maturity, volatility decreases.
- ► The Black-Scholes model with constant volatility does not reflect this process.

Alternatives

- Black model + forward measure (widely used in practice)
- Risk-neutral short-rate models (the theoretically correct way)

The Black Model

Assumptions and input parameters

Assumptions

- Arbitrage-free markets without frictions (transaction costs, taxes etc.)
- European bond option, European caplet, European swaption
- Future value of the underlying is lognormally distributed
- Constant volatility of the underlying (forward price or rate)

Input parameters

- Forward price, forward rate, forward swap rate $F_0(T), f_0(T,T^*), f_0^S(T,T^*)$
- Volatility (standard deviation) of the underlying σ
- Risk-free interest rate r
- Maturity and strike price (rate) of the option $T \quad X, y_X, s_X$

Fischer S. Black
"The Pricing of Commodity Contracts"
Journal of Financial Economics, March 1976

The Black Model

Employing the forward measure

Consider a European call option under the forward measure:

$$C_0 = DF(0,T) \cdot \mathbb{E}^{\mathbb{Q}_T}\left(\max\left(\tilde{P}_T - X, 0\right)\right).$$

We may apply useful substitutions and break up the expectation:*)

$$\mathbb{E}^{\mathbb{Q}_T}\left(\max\left(\tilde{P}_T - X, 0\right)\right) = \underbrace{\mathbb{E}^{\mathbb{Q}_T}\left(\tilde{P}_T\right)}_{\to F_0(T)} \cdot \Phi(d_1) - X \cdot \Phi(d_2).$$

- $F_0(T)$: time-0 price of a forward contract with maturity T

Recall that under the forward measure $\mathbb{Q}_T$, the fair forward price equals the expected future spot price.

$$F_0(T) = \mathbb{E}^{\mathbb{Q}_T}\left(\tilde{P}_T\right)$$

*) As long as $\tilde{F}_T(T)$ is lognormal with $\mathrm{SD}\left[\ln\left(\tilde{F}_T(T)\right)\right] = \sigma_F\sqrt{T}$, we may use a key result from the Black-Scholes (1973) paper to break up the expectation.

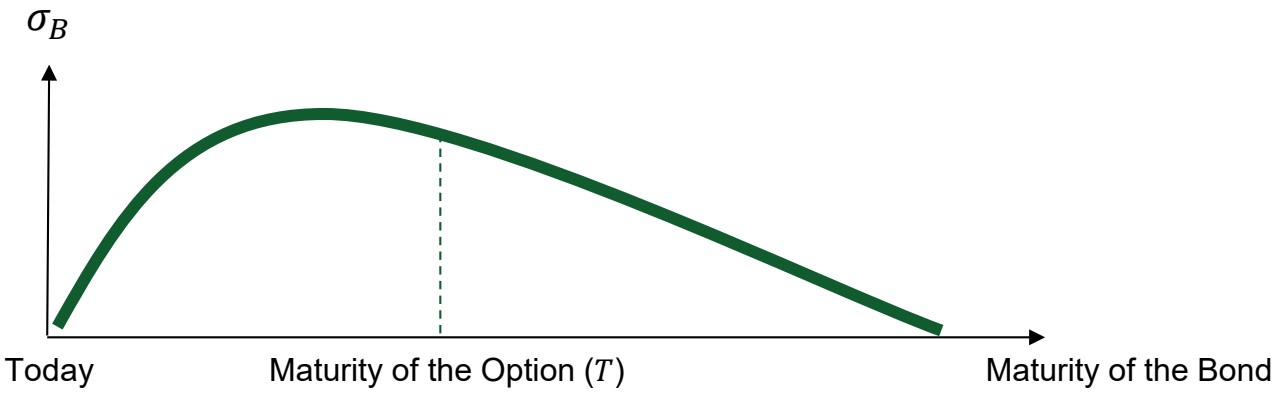
The Black Model

Standard deviation of a (log) bond price

The standard deviation of a (log) bond price is

- zero today, because there is no uncertainty about the price,
- zero at maturity, because we know the bond price will equal the face value.

► Between today and maturity the standard deviation will first increase and then decrease.



The volatility σ_F used in the Black model is proportional to the volatility of the log bond price σ_B at maturity T of the option*)

$$\sigma_F = \frac{\sigma_B}{\sqrt{T}}$$

*) In practice, σ_F is usually derived from forward yield volatilities.

The Black Model

Pricing formula for European bond options

In Black's model, the forward price follows a lognormal process.*)
Under this assumption, the put and call prices of a European bond option are given by

$$C_0 = DF(0,T) \cdot \big(F_0(T)\,\Phi(d_1) - X\,\Phi(d_2)\big) \qquad P_0 = DF(0,T) \cdot \big(X\,\Phi(-d_2) - F_0(T)\,\Phi(-d_1)\big)$$

where

$$d_1 = \frac{\ln\left(\frac{F_0(T)}{X}\right) + \frac{\sigma_F^2}{2}T}{\sigma_F\sqrt{T}} \qquad d_2 = d_1 - \sigma_F\sqrt{T}.$$

- $F_0(T)$: price of the bond forward contract
- σ_F: volatility of the forward returns
- $DF(0,T)$: discount factor for maturity T

Both $F_0(T)$ and P_0 are dirty prices

Φ denotes the standard normal cdf

*) The usual assumption for the process of a forward price in the risk-neutral world is: $\mathrm{d}\tilde{F}_t(T) = \sigma_F\, F_t(T)\, \mathrm{d}\tilde{W}_t$.

The Black Model

Interest rate caps

A *caplet* is a call option on a future interest rate $\tilde{y}_{t_k}(t_{k+1})$ with cap rate y_X and payoff date t_{k+1}.*)
Using Black's model, the price of a caplet (c_0) with notional N is

$$c_0 = N\Delta t_k \cdot DF(0, t_{k+1}) \cdot \left(f_0(t_k, t_{k+1})\, \Phi(d_1) - y_X\, \Phi(d_2)\right)$$

$$d_1 = \frac{\ln\left(\frac{f_0(t_k, t_{k+1})}{y_X}\right) + \frac{\sigma_f^2}{2} t_k}{\sigma_f \sqrt{t_k}} \qquad d_2 = d_1 - \sigma_f \sqrt{t_k}.$$

- $f_0(t_k, t_{k+1})$: forward rate for the future period from t_k to t_{k+1}
- σ_f: volatility of the forward interest rate
- $DF(0, t_{k+1})$: discount factor for maturity t_{k+1}

$\Delta t_k = t_{k+1} - t_k$

Φ denotes the standard normal cdf

*) A cap is a series of caplets with regular strike dates and a final maturity T.

The Black Model

Interest rate floors

A *floorlet* is a put option on a future interest rate $\tilde{y}_{t_k}(t_{k+1})$ with floor rate y_X and payoff date t_{k+1}.*)
Using Black's model, the price of a floorlet (p_0) with notional N is

$$p_0 = N\Delta t_k \cdot DF(0, t_{k+1}) \cdot \left(y_X\, \Phi(-d_2) - f_0(t_k, t_{k+1})\, \Phi(-d_1)\right)$$

$$d_1 = \frac{\ln\left(\frac{f_0(t_k, t_{k+1})}{y_X}\right) + \frac{\sigma_f^2}{2} t_k}{\sigma_f \sqrt{t_k}} \qquad d_2 = d_1 - \sigma_f \sqrt{t_k}.$$

- $f_0(t_k, t_{k+1})$: forward rate for the future period from t_k to t_{k+1}
- σ_f: volatility of the forward interest rate
- $DF(0, t_{k+1})$: discount factor for maturity T

$\Delta t_k = t_{k+1} - t_k$

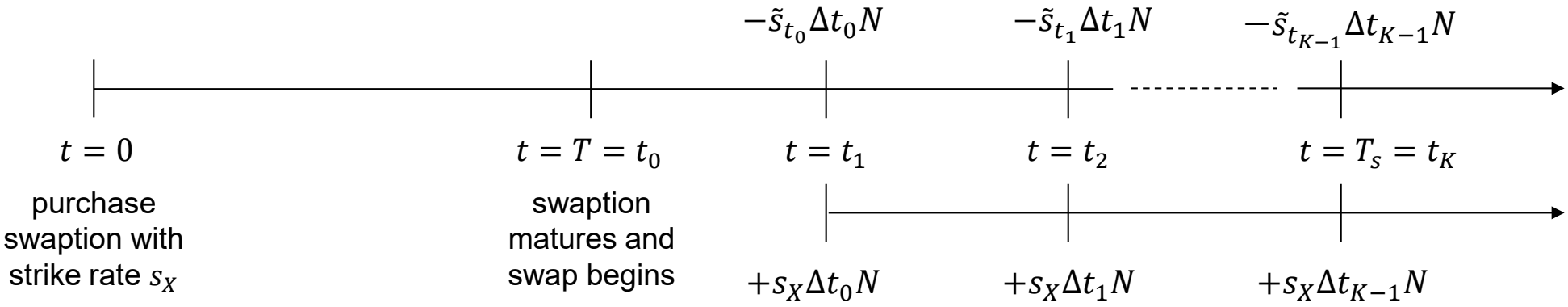
Φ denotes the standard normal cdf

*) A floor is a series of floorlets with regular strike dates and a final maturity T.

The Black Model

Swaptions

A payer *swaption* allows its holder to enter a swap as the payer of a fixed rate s_X at a future date T.
A receiver *swaption* allows its holder to enter a swap as the receiver of a fixed rate s_X at a future date T.



- T is the maturity of the swaption and the start date of the swap
- The t_k $(k = 1 \ldots K)$ are the coupon dates of the swap
- T_S is the maturity of the swap $(T_s = t_K)$

$\Delta t_k = t_{k+1} - t_k$

Illustration is for a receiver swaption

The Black Model

Payer swaptions

Using Black's model, the price of a *payer swaption* with notional N is*)

$$\text{Swaption}_0^p = \sum_{k=1}^{K} N \cdot DF(0, t_k) \cdot \left(f_0^s(T, T_s)\, \Phi(d_1) - s_X\, \Phi(d_2) \right)$$

$$d_1 = \frac{\ln\left(\frac{f_0^s(T, T_s)}{s_X}\right) + \frac{\sigma_s^2}{2} T}{\sigma_s \sqrt{T}} \qquad d_2 = d_1 - \sigma_s \sqrt{T}.$$

- $f_0^s(T, T_s)$: forward swap rate for a swap starting at T and maturing at T_s
- σ_s: volatility of the forward swap rate
- $DF(0, t_k)$: discount factor for period from 0 to t_k

t_k: coupon dates of the swap ($t_K = T_s$)

Φ denotes the standard normal cdf

*) We simplify here by assuming that there is only one swap payment per year. In addition, we ignore day-count conventions.

The Black Model

Receiver swaptions

Using Black's model, the price of a *receiver swaption* with notional N is*)

$$\text{Swaption}_0^r = \sum_{k=1}^{K} N \cdot DF(0, t_k) \cdot \left(s_X \, \Phi(-d_2) - f_0^s(T, T_s) \, \Phi(-d_1)\right)$$

$$d_1 = \frac{\ln\left(\frac{f_0^s(T, T_s)}{s_X}\right) + \frac{\sigma_s^2}{2} T}{\sigma_s \sqrt{T}} \qquad d_2 = d_1 - \sigma_s \sqrt{T}.$$

- $f_0^s(T, T_s)$: forward swap rate for a swap starting at T and maturing at T_s
- σ_s: volatility of the forward swap rate
- $DF(0, t_k)$: discount factor for period from 0 to t_k

t_k: coupon dates of the swap ($t_K = T_s$)

Φ denotes the standard normal cdf

*) We simplify here by assuming that there is only one swap payment per year. In addition, we ignore day-count conventions.

The Black Model

Wrap-up

The Black model for interest rate derivatives takes advantage of the forward measure and assumes that the future value of the underlying is lognormally distributed.

Problems of the Black model

- The different variants (bond options, caps/floors, swaptions) are not consistent.
  - ► When future bond prices are lognormal, future interest rates (and swap rates) are not lognormal.
  - ► When future interest rates are lognormal, future bond prices are not lognormal.
- It cannot be easily extended to value more complex derivatives.
- It does not work for negative (forward) rates (need to add a shift to the forward rate).*)

► More general interest rate risk models fulfill a wider range of purposes.

*) See "Interest Rate Models and Negative Rates" by John Hull.

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